

Math Logic File: Model, Lemmas, and Proofs

Multiple Nuisance Breaks and Spurious Predictability under Persistent Regressors

1 Model Setup and Score Objects

For a scalar array $\mathbf{a} = (a_1, \dots, a_T)'$, write

$$\|\mathbf{a}\|_T^2 = T^{-1} \sum_{t=1}^T a_t^2, \quad \|\mathbf{a}\|_\infty = \max_{t \leq T} |a_t|.$$

Unless stated otherwise, $O_p(\cdot)$ and $o_p(\cdot)$ for growing vectors and matrices are in Euclidean and spectral norm, respectively.

Let

$$\mathbf{Y} = (Y_1, \dots, Y_T)', \quad \mathbf{X}_{lag} = (X_0, \dots, X_{T-1})',$$

and let $\mathbf{w} = (w_1, \dots, w_T)'$, where $w_t = w(X_{t-1})$ is a bounded saturated weight. The predictor satisfies

$$X_t = \theta + \rho_T X_{t-1} + \varepsilon_t. \quad (1.1)$$

The maintained outcome model is

$$Y_t = \beta_0 X_{t-1} + \sum_{j=0}^{q_0} m_j(W_{t-1}) \mathbf{1}\{k_{j,0} < t \leq k_{j+1,0}\} + U_t, \quad (1.2)$$

where

$$0 = k_{0,0} < k_{1,0} < \dots < k_{q_0,0} < k_{q_0+1,0} = T.$$

The coefficient β_0 is stable over the full sample. The structural breaks are confined to the nonparametric nuisance component.

For a candidate partition $\Lambda = (k_1, \dots, k_q)$, write $k_0 = 0$, $k_{q+1} = T$, and

$$I_j(\Lambda) = \{t : k_j < t \leq k_{j+1}\}.$$

The true partition is $\Lambda_0 = (k_{1,0}, \dots, k_{q_0,0})$. Define $j_\Lambda(t) = j$ when $t \in I_j(\Lambda)$. The true nuisance vector is

$$\mu_t = m_{j_{\Lambda_0}(t)}(W_{t-1}), \quad \boldsymbol{\mu} = (\mu_1, \dots, \mu_T)'.$$

Let $\mathbf{P}$ be the $T \times K$ sieve matrix whose t -th row is

$$\mathbf{P}_K(W_{t-1})' = (p_1(W_{t-1}), \dots, p_K(W_{t-1})).$$

For a partition Λ , let $\mathbf{D}_j(\Lambda)$ be the diagonal matrix with diagonal entry $\mathbf{1}\{t \in I_j(\Lambda)\}$. Define

$$\mathbf{Q}_{K,\Lambda} = [\mathbf{D}_0(\Lambda)\mathbf{P}, \mathbf{D}_1(\Lambda)\mathbf{P}, \dots, \mathbf{D}_q(\Lambda)\mathbf{P}], \quad (1.3)$$

and

$$\mathbf{M}_{K,\Lambda} = \mathbf{I}_T - \mathbf{Q}_{K,\Lambda}(\mathbf{Q}_{K,\Lambda}'\mathbf{Q}_{K,\Lambda})^\dagger \mathbf{Q}_{K,\Lambda}'. \quad (1.4)$$

Here † denotes the Moore–Penrose inverse. Thus the definition is valid for every candidate partition; when the block Gram matrix is nonsingular, it coincides with the ordinary inverse. The block nuisance space is

$$\mathcal{N}_{K,\Lambda} = \left\{ \sum_{j=0}^q \mathbf{1}(t \in I_j(\Lambda)) \mathbf{P}_K(W_{t-1})' \mathbf{c}_j : \mathbf{c}_j \in \mathbb{R}^K \right\}.$$

The matrix $\mathbf{M}_{K,\Lambda}$ is the orthogonal residual-maker for this empirical space.

For each β and Λ , define the profiled residual, regime-orthogonalized weight, and scalar score by

$$\hat{\mathbf{u}}(\beta, \Lambda) = \mathbf{M}_{K,\Lambda}(\mathbf{Y} - \beta \mathbf{X}_{lag}), \quad (1.5)$$

$$\mathbf{w}^c(\Lambda) = \mathbf{M}_{K,\Lambda} \mathbf{w}, \quad (1.6)$$

$$\hat{Z}_t(\beta, \Lambda) = \hat{u}_t(\beta, \Lambda) w_t^c(\Lambda). \quad (1.7)$$

The empirical likelihood ratio is

$$L_T(\beta, \Lambda) = \sup \left\{ \prod_{t=1}^T (T\pi_t) : \pi_t \geq 0, \sum_{t=1}^T \pi_t = 1, \sum_{t=1}^T \pi_t \hat{Z}_t(\beta, \Lambda) = 0 \right\}, \quad (1.8)$$

$$\ell_T(\beta, \Lambda) = -2 \log L_T(\beta, \Lambda) = 2 \sum_{t=1}^T \log \{1 + \hat{\lambda}(\beta, \Lambda) \hat{Z}_t(\beta, \Lambda)\}, \quad (1.9)$$

where $\hat{\lambda}(\beta, \Lambda)$ solves the scalar multiplier equation.

All minimizers below are understood as measurable selections with a fixed deterministic tie-breaking rule.

For an interval $[s, e]$, define the null-imposed segment cost

$$\mathcal{C}_K(s, e; \beta) = \min_{\mathbf{c} \in \mathbb{R}^K} \sum_{t=s}^e \{Y_t - \beta X_{t-1} - \mathbf{P}_K(W_{t-1})' \mathbf{c}\}^2. \quad (1.10)$$

For $\Lambda = (k_1, \dots, k_q)$, define

$$RSS_K(\beta, \Lambda) = \sum_{j=0}^q \mathcal{C}_K(k_j + 1, k_{j+1}; \beta). \quad (1.11)$$

Let $\mathcal{L}_q(\epsilon)$ be the set of q -break partitions satisfying minimum spacing. The profile partition estimator is

$$\hat{\Lambda}_q(\beta) = \arg \min_{\Lambda \in \mathcal{L}_q(\epsilon)} RSS_K(\beta, \Lambda). \quad (1.12)$$

If the number of breaks is unknown, define

$$\hat{q}(\beta) = \arg \min_{0 \leq q \leq \bar{q}} \left\{ RSS_K(\beta, \hat{\Lambda}_q(\beta)) + \text{pen}_T(q, K) \right\}, \quad (1.13)$$

and set

$$\hat{\Lambda}(\beta) = \hat{\Lambda}_{\hat{q}(\beta)}(\beta).$$

The break-adaptive statistic is

$$\ell_T^{BA}(\beta) = \ell_T\{\beta, \hat{\Lambda}(\beta)\}. \quad (1.14)$$

2 Assumptions

Assumption 1 (Finite nuisance breaks and smooth regimes). *The number q_0 is fixed. For some $\epsilon > 0$,*

$$\min_{0 \leq j \leq q_0} (k_{j+1,0} - k_{j,0}) \geq \epsilon T.$$

Each m_j belongs to the same smoothness class $\mathcal{H}^s$. If $q_0 \geq 1$, the adjacent breaks satisfy

$$\Delta_T = \min_{0 \leq j < q_0} \|m_{j+1} - m_j\|_{L^2(P_W)} > 0,$$

with $T\Delta_T^2$ large enough relative to the block sieve dimension and logarithmic search complexity. A sufficient fixed-break version is $T\Delta_T^2/(K \log T) \rightarrow \infty$. When $q_0 = 0$, this condition is void.

Assumption 2 (Residualized-weight nondegeneracy). *For the true partition,*

$$T^{-1} \mathbf{w}' \mathbf{M}_{K, \Lambda_0} \mathbf{w} \xrightarrow{p} Q_w, \quad P(Q_w > \underline{q}) = 1$$

for some $\underline{q} > 0$.

Assumption 3 (Block-sieve rates and projection bias). *For the true partition and each regime j , there exists $\mathbf{c}_{j,K}$ such that*

$$m_j(W_{t-1}) = \mathbf{P}_K(W_{t-1})' \mathbf{c}_{j,K} + r_{j,K,t},$$

with $\max_j \|\mathbf{r}_{j,K}\|_T = O_p(a_K)$ and $\max_j \|\mathbf{r}_{j,K}\|_\infty = O_p(a_K)$. Let $\mathbf{r}_0$ denote the stacked approximation-error vector under Λ_0 . The directional projection bias satisfies

$$T^{-1/2} \mathbf{r}_0' \mathbf{M}_{K, \Lambda_0} \mathbf{w} = o_p(1).$$

The block Gram matrix $T^{-1} \mathbf{Q}_{K, \Lambda_0}' \mathbf{Q}_{K, \Lambda_0}$ has eigenvalues bounded away from zero and infinity with probability tending to one. If $\zeta_K = \max_t \|\mathbf{P}_K(W_{t-1})\|$, then

$$a_K(1 + \zeta_K) \rightarrow 0, \quad a_K(1 + \zeta_K)^2 = o(\sqrt{T}),$$

and

$$\frac{(q_0 + 1)K}{\sqrt{T}} \rightarrow 0, \quad \zeta_K \sqrt{\frac{(q_0 + 1)K}{T}} \rightarrow 0.$$

Remark 1 (Projection bias versus undersmoothing). The condition $T^{-1/2} \mathbf{r}_0' \mathbf{M}_{K, \Lambda_0} \mathbf{w} = o_p(1)$ only requires the nuisance approximation error to be negligible in the residualized score direction. It is weaker than the full root- T undersmoothing condition $\sqrt{T} a_K \rightarrow 0$. A sufficient primitive route is a product-rate condition $\sqrt{T} a_{m,K} a_{w,K} \rightarrow 0$, where $a_{m,K}$ approximates the nuisance functions and $a_{w,K}$ approximates the conditional mean of the weight w_t given W_{t-1} .

Assumption 4 (Known-partition oracle score). With $\mathbf{v}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{w}$, define $\tilde{\mathbf{U}}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{U}$ and

$$A_T^{(q)} = T^{-1/2} \sum_{t=1}^T U_t v_{t,0}, \quad V_T^{(q)} = T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2.$$

There exist a random scale η_q , a constant $c_\eta > 0$, and a standard normal Z , independent of the limiting sigma-field generating η_q , such that

$$A_T^{(q)} \xrightarrow{\text{st}} \eta_q Z, \quad V_T^{(q)} \xrightarrow{p} \eta_q^2, \quad P(\eta_q \geq \sqrt{c_\eta}) = 1.$$

In addition,

$$\max_{t \leq T} |U_t v_{t,0}| = o_p(\sqrt{T}),$$

$$T^{-1} \sum_{t=1}^T \{(\tilde{U}_{0,t} v_{t,0})^2 - (U_t v_{t,0})^2\} = o_p(1), \quad \max_{t \leq T} |(\tilde{U}_{0,t} - U_t) v_{t,0}| = o_p(\sqrt{T}).$$

Finally, the feasible true-partition score array satisfies the scalar empirical-likelihood convex-hull condition

$$P \left\{ \min_{t \leq T} \hat{Z}_t(\beta_0, \Lambda_0) < 0 < \max_{t \leq T} \hat{Z}_t(\beta_0, \Lambda_0) \right\} \rightarrow 1.$$

This condition is imposed on the feasible array $\hat{Z}_t(\beta_0, \Lambda_0)$, not only on the oracle array $U_t v_{t,0}$.

For the fixed-order analysis, define

$$\tilde{\Lambda} = \hat{\Lambda}_{q_0}(\beta_0), \quad \tilde{\Lambda} = (\tilde{k}_1, \dots, \tilde{k}_{q_0}).$$

The unknown-order estimator $\hat{\Lambda}(\beta_0)$ is used only in Corollary 1.

Assumption 5 (Fixed-order partition localization). For the true number of breaks,

$$\max_{1 \leq j \leq q_0} |\tilde{k}_j - k_{j,0}| = O_p(r_T), \quad r_T = o(\sqrt{T}).$$

Equivalently, the number of fixed-order observations assigned to the wrong regime satisfies

$$|\mathcal{M}_T| = |\{t : j_{\tilde{\Lambda}}(t) \neq j_{\Lambda_0}(t)\}| = o_p(\sqrt{T}).$$

This localization condition does not by itself imply score equivalence, because moving boundary observations also perturbs the fitted regime coefficients outside $\mathcal{M}_T$.

Assumption 6 (High-level fixed-order estimated-score equivalence). Under $H_0 : \beta = \beta_0$,

$$T^{-1/2} \sum_{t=1}^T \{\hat{Z}_t(\beta_0, \tilde{\Lambda}) - \hat{Z}_t(\beta_0, \Lambda_0)\} = o_p(1), \quad (2.1)$$

$$T^{-1} \sum_{t=1}^T \{\hat{Z}_t(\beta_0, \tilde{\Lambda})^2 - \hat{Z}_t(\beta_0, \Lambda_0)^2\} = o_p(1), \quad (2.2)$$

$$\max_{t \leq T} |\hat{Z}_t(\beta_0, \tilde{\Lambda})| = o_p(\sqrt{T}). \quad (2.3)$$

Zero lies in the interior of the scalar empirical-likelihood convex hull for the fixed-order estimated-partition score array with probability tending to one. This is a high-level score-equivalence condition; Assumption 5 alone is not claimed to prove it.

Remark 2 (Estimated-partition bridge as the main proof risk). The critical gap between the oracle Wilks theorem and the feasible break-adaptive statistic is not the finite-sample projection algebra. It is the bridge

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = o_p(1), \quad V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = o_p(1),$$

where $S_T(\Lambda)$ and $V_T(\Lambda)$ are defined below. Assumption 5 controls how many boundary observations are assigned to the wrong regime, but it does not by itself control the change in fitted block coefficients away from the boundary. Assumption 6 is therefore kept as a high-level bridge unless a separate perturbation argument establishes these two displays.

3 Statements and Proofs

The stable no-break score is

$$S_T^{\text{st}}(\beta_0) = T^{-1/2}(\mathbf{Y} - \beta_0 \mathbf{X}_{\text{lag}})' \mathbf{M}_{K,0} \mathbf{w},$$

where 0 denotes the no-break partition.

Proposition 1 (Stable-score decomposition). *Under (1.2), the stable-nuisance score satisfies the exact identity*

$$S_T^{\text{st}}(\beta_0) = A_T^{\text{st}} + B_T,$$

where

$$A_T^{\text{st}} = T^{-1/2} \mathbf{U}' \mathbf{M}_{K,0} \mathbf{w}, \quad B_T = T^{-1/2} \boldsymbol{\mu}' \mathbf{M}_{K,0} \mathbf{w}.$$

The term B_T is the stable-projection nuisance component left by imposing the no-break nuisance projection. When $q_0 = 0$, this component is asymptotically negligible under the projection-bias condition in Assumption 3; when $q_0 > 0$, a nonnegligible component is interpreted as break-induced.

Proof. Under $H_0 : \beta = \beta_0$,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{\text{lag}} = \boldsymbol{\mu} + \mathbf{U}.$$

Multiplying by the stable residualized weight gives

$$T^{-1/2}(\mathbf{Y} - \beta_0 \mathbf{X}_{\text{lag}})' \mathbf{M}_{K,0} \mathbf{w} = T^{-1/2} \mathbf{U}' \mathbf{M}_{K,0} \mathbf{w} + T^{-1/2} \boldsymbol{\mu}' \mathbf{M}_{K,0} \mathbf{w}.$$

This is an exact algebraic identity. No stable-sieve approximation remainder is needed because B_T is defined using the exact nuisance vector $\boldsymbol{\mu}$.

When $q_0 = 0$, write $\boldsymbol{\mu} = \mathbf{P} \mathbf{c}_K + \mathbf{r}_K$. Since $\mathbf{M}_{K,0} \mathbf{P} \mathbf{c}_K = \mathbf{0}$,

$$B_T = T^{-1/2} \mathbf{r}_K' \mathbf{M}_{K,0} \mathbf{w} = o_p(1)$$

by the projection-bias condition in Assumption 3 with $\Lambda_0 = 0$. □

Definition 1 (Break-induced spurious predictivity). The stable nuisance specification generates break-induced spurious predictivity at β_0 if

$$T^{-1/2} \langle \mathbf{M}_{K,0} \boldsymbol{\mu}, \mathbf{M}_{K,0} \mathbf{w} \rangle_{\mathbb{R}^T} = B_T \neq o_p(1).$$

The condition is directional: a nuisance break with little projection on the residualized persistent weight need not distort the test.

For the stable no-break empirical-likelihood statistic, write

$$V_T^{\text{st}} = T^{-1} \sum_{t=1}^T \widehat{Z}_t(\beta_0, 0)^2,$$

where $\widehat{Z}_t(\beta_0, 0)$ is the score in (1.7) computed under the no-break partition.

Theorem 1 (Omitted-break failure). *Suppose the stable empirical-likelihood statistic admits*

$$\ell_T^{\text{st}}(\beta_0) = \frac{\{S_T^{\text{st}}(\beta_0)\}^2}{V_T^{\text{st}}} + o_p(1),$$

with V_T^{st} bounded away from zero in probability. Define

$$a_T = \frac{A_T^{\text{st}}}{(V_T^{\text{st}})^{1/2}}, \quad b_T = \frac{B_T}{(V_T^{\text{st}})^{1/2}}.$$

If $a_T = O_p(1)$ and $|b_T| \rightarrow_p \infty$, then, for every fixed chi-square critical value c_α ,

$$P\{\ell_T^{\text{st}}(\beta_0) > c_\alpha\} \rightarrow 1.$$

If instead

$$(a_T, b_T) \xrightarrow{\text{st}} (Z, B),$$

where $Z \sim N(0, 1)$ is independent of the limiting sigma-field and B is measurable with respect to that sigma-field, then

$$\ell_T^{\text{st}}(\beta_0) \Rightarrow (Z + B)^2.$$

Conditionally on B , the limit is $\chi_1^2(B^2)$. Thus a constant $B = \delta$ gives $\chi_1^2(\delta^2)$, a random B gives a mixture of noncentral chi-squares, and $B = 0$ almost surely gives the central χ_1^2 limit.

Proof. By Proposition 1,

$$S_T^{\text{st}}(\beta_0) = A_T^{\text{st}} + B_T.$$

Hence the scalar expansion is $\ell_T^{\text{st}}(\beta_0) = (a_T + b_T)^2 + o_p(1)$. If $a_T = O_p(1)$ and $|b_T| \rightarrow_p \infty$, then

$$|a_T + b_T| \geq |b_T| - |a_T| \rightarrow_p \infty,$$

so the statistic diverges in probability. Under joint stable convergence, the continuous mapping theorem gives $(a_T + b_T)^2 \Rightarrow (Z + B)^2$. Since Z is conditionally standard normal given the limiting sigma-field, $(Z + B)^2 \mid B \sim \chi_1^2(B^2)$. $\square$

Remark 3 (Scope of omitted-break divergence). The divergence part of Theorem 1 is conditional on the displayed stable empirical-likelihood quadratic expansion. With a fixed omitted nuisance break, B_T can be of order $\sqrt{T}$, and the null-style scalar expansion need not follow automatically. Establishing generic fixed-break divergence therefore requires a separate expansion under the misspecified stable nuisance projection.

Lemma 1 (Block projection identities). *For any partition Λ ,*

$$\mathbf{Q}'_{K,\Lambda} \mathbf{M}_{K,\Lambda} = \mathbf{0}, \quad \mathbf{M}'_{K,\Lambda} = \mathbf{M}_{K,\Lambda}, \quad \mathbf{M}_{K,\Lambda}^2 = \mathbf{M}_{K,\Lambda}.$$

For every $\mathbf{c}$ conformable with $\mathbf{Q}_{K,\Lambda}$,

$$(\mathbf{Q}_{K,\Lambda} \mathbf{c})' \mathbf{M}_{K,\Lambda} \mathbf{w} = 0.$$

Proof. Let

$$\mathbf{\Pi}_{K,\Lambda} = \mathbf{Q}_{K,\Lambda} (\mathbf{Q}'_{K,\Lambda} \mathbf{Q}_{K,\Lambda})^\dagger \mathbf{Q}'_{K,\Lambda}.$$

This is the orthogonal projector onto the column space of $\mathbf{Q}_{K,\Lambda}$. Hence $\mathbf{\Pi}'_{K,\Lambda} = \mathbf{\Pi}_{K,\Lambda}$, $\mathbf{\Pi}_{K,\Lambda}^2 = \mathbf{\Pi}_{K,\Lambda}$, and $\mathbf{Q}'_{K,\Lambda} (\mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda}) = \mathbf{0}$. Since $\mathbf{M}_{K,\Lambda} = \mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda}$, the displayed identities follow. The final equality is obtained by multiplying $\mathbf{Q}'_{K,\Lambda} \mathbf{M}_{K,\Lambda} = \mathbf{0}$ by $\mathbf{c}$ and $\mathbf{w}$. $\square$

Set $\Lambda = \Lambda_0$ and write

$$\mathbf{v}_0 = \mathbf{M}_{K,\Lambda_0} \mathbf{w}, \quad \hat{\mathbf{u}}_0(\beta_0) = \mathbf{M}_{K,\Lambda_0} (\mathbf{Y} - \beta_0 \mathbf{X}_{lag}).$$

Under the true partition,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \mathbf{Q}_{K,\Lambda_0} \mathbf{c}_0 + \mathbf{r}_0 + \mathbf{U},$$

where $\mathbf{r}_0$ stacks the regime-specific sieve approximation errors. Let $\mathbf{e}_0 = \mathbf{M}_{K,\Lambda_0} \mathbf{r}_0$.

Lemma 2 (Known-partition replacement). *Under Assumptions 3 and 4,*

$$T^{-1/2} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^T U_t v_{t,0} + o_p(1), \quad (3.1)$$

$$T^{-1} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda_0)^2 = T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2 + o_p(1), \quad (3.2)$$

$$\max_{t \leq T} |\hat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}). \quad (3.3)$$

Proof. Using Lemma 1,

$$\hat{\mathbf{u}}_0(\beta_0) = \mathbf{M}_{K,\Lambda_0} (\mathbf{U} + \mathbf{r}_0), \quad \mathbf{v}_0 = \mathbf{M}_{K,\Lambda_0} \mathbf{w}.$$

Because $\mathbf{M}_{K,\Lambda_0}$ is symmetric and idempotent,

$$T^{-1/2} \hat{\mathbf{u}}_0(\beta_0)' \mathbf{v}_0 = T^{-1/2} (\mathbf{U} + \mathbf{r}_0)' \mathbf{v}_0.$$

By Assumption 3,

$$T^{-1/2} \mathbf{r}_0' \mathbf{v}_0 = T^{-1/2} \mathbf{r}_0' \mathbf{M}_{K,\Lambda_0} \mathbf{w} = o_p(1),$$

which proves (3.1).

It remains to record the denominator and maximum bounds. Let $G_T = T^{-1} \mathbf{Q}'_{K,\Lambda_0} \mathbf{Q}_{K,\Lambda_0}$. Since the rows of $\mathbf{Q}_{K,\Lambda_0}$ have norm at most ζ_K and the eigenvalues of G_T are bounded away from zero and infinity,

$$\|\mathbf{v}_0\|_\infty = O_p(1 + \zeta_K).$$

Moreover,

$$T^{-1} \mathbf{Q}'_{K, \Lambda_0} \mathbf{r}_0 = O_p(a_K),$$

and hence

$$\|\mathbf{e}_0\|_T \leq \|\mathbf{r}_0\|_T = O_p(a_K), \quad \|\mathbf{e}_0\|_\infty = O_p\{a_K(1 + \zeta_K)\}.$$

The stated rates imply

$$\|\mathbf{e}_0 \odot \mathbf{v}_0\|_T = O_p\{a_K(1 + \zeta_K)\} = o_p(1), \quad \|\mathbf{e}_0 \odot \mathbf{v}_0\|_\infty = O_p\{a_K(1 + \zeta_K)^2\} = o_p(\sqrt{T}).$$

For the denominator, write $\tilde{\mathbf{U}}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{U}$. Then

$$\hat{Z}_t(\beta_0, \Lambda_0) = \tilde{U}_{0,t} v_{t,0} + e_{0,t} v_{t,0}.$$

Assumption 4 gives

$$T^{-1} \sum_{t=1}^T \{(\tilde{U}_{0,t} v_{t,0})^2 - (U_t v_{t,0})^2\} = o_p(1),$$

and Cauchy–Schwarz with $\|\mathbf{e}_0 \odot \mathbf{v}_0\|_T = o_p(1)$ gives the remaining cross and square terms. This proves (3.2). The maximum bound follows from

$$|\hat{Z}_t(\beta_0, \Lambda_0)| \leq |U_t v_{t,0}| + |(\tilde{U}_{0,t} - U_t) v_{t,0}| + |e_{0,t} v_{t,0}|,$$

Assumption 4, and $\|\mathbf{e}_0 \odot \mathbf{v}_0\|_\infty = o_p(\sqrt{T})$. $\square$

Lemma 3 (Scalar empirical-likelihood expansion). *Let $Z_{t,T}$ be scalar scores and $\hat{V}_T^Z = T^{-1} \sum_t Z_{t,T}^2$. Suppose zero lies in the interior of the scalar convex hull with probability tending to one and*

$$T^{-1/2} \sum_t Z_{t,T} = O_p(1), \quad \hat{V}_T^Z \rightarrow_p \eta^2, \quad \max_t |Z_{t,T}| = o_p(\sqrt{T}),$$

where $P(\eta^2 \geq c) = 1$ for some fixed $c > 0$. Then

$$2 \sum_{t=1}^T \log(1 + \hat{\lambda} Z_{t,T}) = \frac{\{T^{-1/2} \sum_t Z_{t,T}\}^2}{T^{-1} \sum_t Z_{t,T}^2} + o_p(1).$$

Proof. The second-moment condition implies $P(\hat{V}_T^Z \geq c/2) \rightarrow 1$. Let $S_T^Z = \sum_t Z_{t,T}$ and $Q_T^Z = \sum_t Z_{t,T}^2$. On this event, $S_T^Z = O_p(\sqrt{T})$ and $Q_T^Z \geq cT/2$. Monotonicity of the scalar multiplier equation, evaluated at $\pm 2|S_T^Z|/Q_T^Z$, gives

$$|\hat{\lambda}| \leq 2|S_T^Z|/Q_T^Z = O_p(T^{-1/2}).$$

Since $\max_t |\hat{\lambda} Z_{t,T}| = o_p(1)$, a Taylor expansion of the multiplier equation gives

$$\hat{\lambda} = \frac{\sum_t Z_{t,T}}{\sum_t Z_{t,T}^2} + o_p(T^{-1/2}).$$

Substitution into the second-order expansion of the log empirical likelihood gives the displayed quadratic approximation. $\square$

Theorem 2 (Known-partition Wilks theorem). *Under Assumptions 1, 2, 3, and 4, under $H_0 : \beta = \beta_0$,*

$$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2.$$

Proof. Lemma 2 reduces the feasible score to the matched block-projection oracle score $U_t v_{t,0}$. Assumption 4 gives

$$A_T^{(q)} \xrightarrow{\text{st}} \eta_q Z, \quad V_T^{(q)} \rightarrow_p \eta_q^2, \quad P(\eta_q \geq \sqrt{c_\eta}) = 1.$$

Stable Slutsky therefore gives $A_T^{(q)} / \sqrt{V_T^{(q)}} \Rightarrow Z$. Lemma 3 yields

$$\ell_T(\beta_0, \Lambda_0) = \frac{\{T^{-1/2} \sum_t U_t v_{t,0}\}^2}{T^{-1} \sum_t U_t^2 v_{t,0}^2} + o_p(1) \Rightarrow Z^2 \sim \chi_1^2.$$

□

$$S_T(\Lambda) = T^{-1/2} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda), \quad V_T(\Lambda) = T^{-1} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda)^2.$$

By symmetry and idempotence of $\mathbf{M}_{K,\Lambda}$,

$$S_T(\Lambda) = T^{-1/2} (\mathbf{Y} - \beta_0 \mathbf{X}_{lag})' \mathbf{M}_{K,\Lambda} \mathbf{w}. \quad (3.4)$$

Lemma 4 (Partition perturbation implies score perturbation: crude route). *Let*

$$\Delta_M = \mathbf{M}_{K,\tilde{\Lambda}} - \mathbf{M}_{K,\Lambda_0}.$$

Consider the fixed-order case with the correct number of breaks and

$$\max_j |\tilde{k}_j - k_{j,0}| = O_p(r_T), \quad r_T = o(\sqrt{T}).$$

Suppose bounded weights, bounded nuisance functions, bounded basis envelope ζ_K , strong minimum spacing, and uniform Gram stability over the r_T -neighborhood of Λ_0 imply the perturbation bounds

$$T^{-1/2} \mathbf{U}' \Delta_M \mathbf{w} = O_p\left(\zeta_K \sqrt{\frac{K r_T}{T}}\right) + o_p(1), \quad (3.5)$$

$$T^{-1/2} \boldsymbol{\mu}' \Delta_M \mathbf{w} = O_p\left(\frac{(1 + \zeta_K) r_T}{\sqrt{T}}\right) + o_p(1), \quad (3.6)$$

$$V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = O_p\left(\frac{(1 + \zeta_K)^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1), \quad (3.7)$$

$$\max_{t \leq T} |\hat{Z}_t(\beta_0, \tilde{\Lambda})| = o_p(\sqrt{T}). \quad (3.8)$$

If

$$\zeta_K \sqrt{\frac{K r_T}{T}} \rightarrow 0, \quad \frac{(1 + \zeta_K) r_T}{\sqrt{T}} \rightarrow 0, \quad \frac{\zeta_K^2 K r_T}{T} \rightarrow 0,$$

then the numerator, denominator, and maximum-score parts of Assumption 6 hold. If the estimated-partition convex-hull condition also holds, then the full fixed-order estimated-score equivalence assumption holds.

Proof. Under $H_0 : \beta = \beta_0$, $\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \boldsymbol{\mu} + \mathbf{U}$. Using (3.4),

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = T^{-1/2}(\mathbf{U} + \boldsymbol{\mu})' \Delta_M \mathbf{w}.$$

The two numerator bounds (3.5) and (3.6), together with the displayed rates, give

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = o_p(1).$$

The denominator requirement is (3.7). The maximum-score requirement is (3.8). The estimated-partition convex-hull condition must still be verified separately or imposed as part of Assumption 6. $\square$

Remark 4 (How to use Lemma 4). This is intentionally a crude de-risking lemma. Its role is to isolate the proof target for a primitive one-break, known- q_0 , strong-spacing analysis. Boundary observations should generate the $r_T/\sqrt{T}$ nuisance term; least-squares coefficient perturbation should generate the $\zeta_K \sqrt{K r_T/T}$ stochastic term; and the denominator should be smaller after T^{-1} normalization. Proving those perturbation bounds in the simplest bounded-basis case would turn the estimated-partition bridge from a purely high-level assumption into a primitive sufficient condition.

Lemma 5 (Fixed-order estimated-partition likelihood transfer). *Under the assumptions of Theorem 2 and Assumption 6,*

$$\ell_T(\beta_0, \tilde{\Lambda}) - \ell_T(\beta_0, \Lambda_0) = o_p(1).$$

Proof. Lemma 2 and Assumption 4 give $S_T(\Lambda_0) = O_p(1)$, $V_T(\Lambda_0) \rightarrow_p \eta_q^2$, and a true-partition maximum-score bound. Assumption 6 gives

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = o_p(1), \quad V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = o_p(1),$$

and the fixed-order estimated-partition maximum-score and convex-hull conditions. Lemma 3 applies to both arrays, and the two quadratic ratios differ by $o_p(1)$ because both denominators are bounded away from zero with probability tending to one. $\square$

Theorem 3 (Fixed-order estimated-partition Wilks theorem). *Under the assumptions of Theorem 2 and Assumption 6,*

$$\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2.$$

This theorem is high-level: it proves that fixed-order estimated-score equivalence implies empirical-likelihood equivalence, not that the profile-sieve partition estimator satisfies Assumption 6 from primitive conditions.

Proof. By Lemma 5,

$$\ell_T(\beta_0, \tilde{\Lambda}) = \ell_T(\beta_0, \Lambda_0) + o_p(1).$$

The result follows from Theorem 2. $\square$

Corollary 1 (Unknown number of breaks). *If $P\{\hat{q}(\beta_0) = q_0\} \rightarrow 1$ and Assumption 6 holds on the fixed-order event, then*

$$\ell_T^{BA}(\beta_0) \Rightarrow \chi_1^2.$$

Proof. On the event $\{\widehat{q}(\beta_0) = q_0\}$, $\widehat{\Lambda}(\beta_0) = \widetilde{\Lambda}$. Since this event has probability tending to one, Theorem 3 gives the same Wilks limit for $\ell_T^{BA}(\beta_0)$. $\square$

Theorem 4 (Local predictive alternatives). *Let $\beta_T = \beta_0 + d_T$, where $d_T \rightarrow 0$. Maintain Assumptions 1, 2, and 3. In addition, assume that the oracle-score part of Assumption 4 continues to hold under the local law P_{β_T} :*

$$T^{-1/2} \sum_{t=1}^T U_t v_{t,0} \xrightarrow{\text{st}} \eta_q Z, \quad T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2, \quad \max_{t \leq T} |U_t v_{t,0}| = o_p(\sqrt{T}),$$

with $P(\eta_q \geq \sqrt{c_\eta}) = 1$. Define

$$g_{t,T} = d_T(\mathbf{M}_{K,\Lambda_0} \mathbf{X}_{lag})_t v_{t,0}.$$

Suppose

$$T^{-1/2} \sum_{t=1}^T g_{t,T} \xrightarrow{p} \eta_q \delta_h, \tag{3.9}$$

$$T^{-1} \sum_{t=1}^T g_{t,T}^2 = o_p(1), \tag{3.10}$$

$$\max_{t \leq T} |g_{t,T}| = o_p(\sqrt{T}), \tag{3.11}$$

with δ_h fixed, and set $Z_{t,T}^{\text{loc}} = U_t v_{t,0} + g_{t,T}$. Under the local data-generating law β_T , assume the feasible true-partition score satisfies

$$T^{-1/2} \sum_{t=1}^T \{\widehat{Z}_t(\beta_0, \Lambda_0) - Z_{t,T}^{\text{loc}}\} = o_p(1), \tag{3.12}$$

$$T^{-1} \sum_{t=1}^T \{\widehat{Z}_t(\beta_0, \Lambda_0)^2 - (Z_{t,T}^{\text{loc}})^2\} = o_p(1), \tag{3.13}$$

$$\max_{t \leq T} |\widehat{Z}_t(\beta_0, \Lambda_0) - Z_{t,T}^{\text{loc}}| = o_p(\sqrt{T}). \tag{3.14}$$

Also assume the feasible true-partition convex-hull condition holds under β_T :

$$P_{\beta_T} \left\{ \min_{t \leq T} \widehat{Z}_t(\beta_0, \Lambda_0) < 0 < \max_{t \leq T} \widehat{Z}_t(\beta_0, \Lambda_0) \right\} \rightarrow 1.$$

Assume, in addition, that the fixed-order estimated-score equivalence in Assumption 6 holds under β_T , with all o_p statements evaluated under the local law. Then, when the statistic is evaluated at β_0 ,

$$\ell_T(\beta_0, \widetilde{\Lambda}) \Rightarrow \chi_1^2(\delta_h^2).$$

If $P\{\widehat{q}(\beta_0) = q_0\} \rightarrow 1$, the same limit holds for $\ell_T^{BA}(\beta_0)$.

Proof. Under the local alternative,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \boldsymbol{\mu} + \mathbf{U} + d_T \mathbf{X}_{lag}.$$

At the true partition, the block projection removes the regime-specific sieve approximation to $\boldsymbol{\mu}$. The numerator replacement condition (3.12) gives

$$T^{-1/2} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^T U_t v_{t,0} + T^{-1/2} \sum_{t=1}^T g_{t,T} + o_p(1).$$

By (3.9), the second term converges to $\eta_q \delta_h$. The denominator replacement condition (3.13) reduces the feasible denominator to the shifted oracle denominator. Moreover,

$$T^{-1} \sum_t (U_t v_{t,0} + g_{t,T})^2 = V_T^{(q)} + 2T^{-1} \sum_t U_t v_{t,0} g_{t,T} + T^{-1} \sum_t g_{t,T}^2.$$

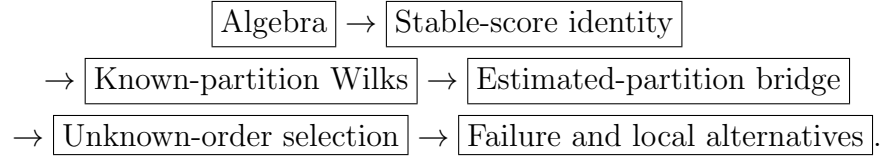
The last term is $o_p(1)$ by (3.10); the cross term is $o_p(1)$ by Cauchy–Schwarz and $V_T^{(q)} = O_p(1)$. Thus the feasible denominator remains $\eta_q^2 + o_p(1)$. The maximum condition follows from the oracle maximum condition, (3.11), and (3.14). The feasible convex-hull condition and Lemma 3 give the known-partition limit $(Z + \delta_h)^2$. Lemma 5, applied under the local-alternative estimated-score and convex-hull conditions, transfers the result to $\widetilde{\Lambda}$. If $\widehat{q}(\beta_0) = q_0$ with probability tending to one, then $\widehat{\Lambda}(\beta_0) = \widetilde{\Lambda}$ on that event, giving the unknown-order statement. $\square$

4 Proof Dependency Map and Risk Register

This file is a math-spine document. It keeps only the model, assumptions, definitions, lemmas, theorems, proofs, and dependency checks needed to evaluate whether

$$\text{assumptions} \implies \text{Wilks and local-power conclusions.}$$

The intended audit order is



Dependency map

Result	Direct argument in this file	High-level or remaining input
Projection algebra	Lemma 1; Moore–Penrose projection gives $\mathbf{M}' = \mathbf{M}$, $\mathbf{M}^2 = \mathbf{M}$, and $\mathbf{Q}'\mathbf{M} = \mathbf{0}$.	None beyond conformability of $\mathbf{Q}_{K,\Lambda}$.
Stable-score decomposition	Proposition 1; exact identity $S_T^{\text{st}} = T^{-1/2}\mathbf{U}'\mathbf{M}_{K,0}\mathbf{w} + T^{-1/2}\boldsymbol{\mu}'\mathbf{M}_{K,0}\mathbf{w}$.	Do not overclaim fixed-break divergence without the stable EL expansion in Theorem 1.
Known-partition Wilks	Lemmas 2 and 3, then Theorem 2.	Oracle mixed-normal score, feasible convex hull, max condition, and projection-noise replacement in Assumption 4.
Estimated fixed-order partition	Lemma 5 transfers EL once numerator, denominator, max, and convex-hull equivalence hold.	Assumption 6; Lemma 4 records a crude route to primitive sufficient conditions.
Unknown number of breaks	Corollary 1; reduce to fixed-order theorem on $\{\widehat{q}(\beta_0) = q_0\}$.	Order-selection consistency $P\{\widehat{q}(\beta_0) = q_0\} \rightarrow 1$.
Local alternatives	Theorem 4; shifted score $U_tv_{t,0} + g_{t,T}$ gives $\chi_1^2(\delta_h^2)$.	Oracle score limit under the local law, local shifted-array replacement, convex hull, and estimated-score equivalence.

Risk register

Component	Current status	Risk	Next mathematical action
Projection algebra	Exact finite-sample result.	Low	Keep the Moore–Penrose formulation so candidate singular Gram matrices are covered.
Stable-score decomposition	Exact identity.	Low	Maintain the distinction between score contamination and automatic fixed-break EL divergence.
Known-partition Wilks	Conditional theorem is coherent.	Medium	Verify the oracle CLT, denominator, full-sample projection, and max condition from primitive persistence assumptions.
Estimated-partition bridge	High-level transfer plus crude perturbation target.	High	Prove Lemma 4’s bounds first for one break, known q_0 , bounded basis, strong spacing, and martingale or i.i.d. errors.
Unknown q	Conditional corollary.	Medium–high	Keep $P\{\widehat{q}(\beta_0) = q_0\} \rightarrow 1$ as a high-level condition until the fixed-order bridge is de-risked.
Omitted-break failure	Conditional on stable EL expansion.	Medium	Do not state generic fixed-break divergence without a misspecified stable-EL expansion.
Local alternatives	Conditional shifted-score theorem.	Medium	Check the local-law oracle score and replacement relative to $U_t v_{t,0} + g_{t,T}$, not only relative to $U_t v_{t,0}$.

Exit criteria for paper polish

The theory is ready to support story-level rewriting only after the following three questions have precise answers:

1. Why does the known-partition statistic have a χ_1^2 limit?
2. Why does replacing Λ_0 by $\widetilde{\Lambda}$ not change the score to first order?
3. Is unknown q_0 proved from primitive penalty conditions, or is it imposed through $P\{\widehat{q}(\beta_0) = q_0\} \rightarrow 1$?